

Department of Computer Science & Engineering(Networks)

6CSN-1 Mini project Title & Abstract Submission Form [A.Y:2025-26]

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Seminar Title:

Design and Implementation of a Customized AI Model using Transfer Learning and Fine-Tuning

Abstract:

This project focuses on designing and implementing a customized artificial intelligence model by adapting pre-trained models using transfer learning and fine-tuning techniques. The system leverages powerful frameworks such as TensorFlow or PyTorch to modify existing models for specific tasks and datasets. By reusing learned features from large-scale models, the approach significantly reduces training time and computational cost while improving performance and accuracy. Fine-tuning is applied to adjust model parameters according to domain-specific requirements, enabling better prediction and classification results. This project demonstrates the practical application of modern AI techniques to build efficient, scalable, and high-performance models suitable for real-world problems in areas such as image recognition, natural language processing, and predictive analytics.

Benefit to Current Society/Scenario:

The development of a customized AI model using transfer learning and fine-tuning provides significant benefits to modern society by enabling faster and more efficient deployment of intelligent systems. By leveraging frameworks like TensorFlow and PyTorch, organizations can build accurate and domain-specific solutions without the need for large datasets or high computational resources. This approach reduces development time and cost, making AI more accessible to startups and small businesses. It supports applications in healthcare, education, finance, and industry by improving automation, decision-making, and prediction accuracy. Overall, it accelerates innovation and helps industries adopt AI-driven solutions effectively in today's data-driven world.

Counsellor Signature

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